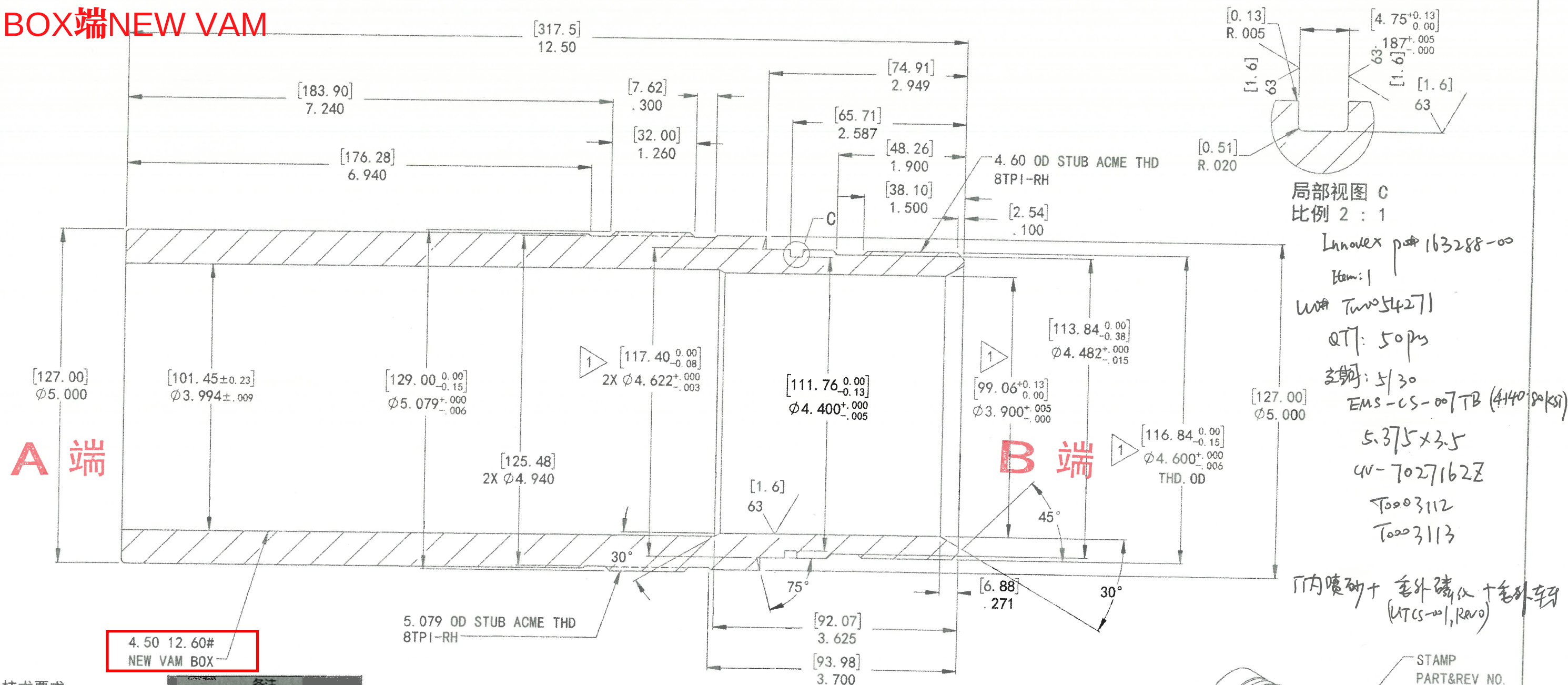


8121005465-10:
BOX端NEW VAM



技术要求

1. 去毛刺
2. ACME螺纹处喷砂
3. 磷化处理

4.50 12.60#
NEW VAM BOX

备注
1. 每个VAM螺纹端面会车掉0.25mm;
2. 预留VAM螺纹重切长度;
3. VAM CZ只对VAM螺纹负责, 其它尺寸、外形等概不负责。

1 THESE DIMENSIONS TO BE CONCENTRIC WITHIN .005 TIR. 这些尺寸的同轴度在 .005 [0.13] 以内

Unit in [X] is mm.

						TOLERANCE FOR MACHINED SURFACES UNLESS OTHERWISE SPECIFIED .X" = ± .06" X (mm) = ± 1.52mm .XX" = ± .03" X (mm) = ± 0.76mm .XXX" = ± .015" .X X (mm) = ± 0.38mm X° = ± 2° X° = ± 2°	TITLE: TOP SUB MATERIAL: L80 22HRC 80KSI DATE: 2017/11/15 DRAWN BY: JIM CHECKED: 李富全 APPROVED: 张衡	DRAWING/MATERIAL NO.: 314-4500-C1-0014 HEAT TREAT: N/A PRODUCT FAMILY: 314 COATING: UTCS-001	REV: 02 PAGE: 1/1
	Added concentricity requirements	2024. 11. 14	李富全			ANGLES TO BE 30° FROM HORIZONTAL CONCENTRICITY .015" [0.38mm] TIR ALL EDGES .015"x45° [0.38mmx45°] INSIDE CORNERS .03"-.06" [0.8-1.5mm] R SURFACE FINISH MAX. THREADS MAX.			
	Updated the coating	2020. 7. 28	YC						
NO.	REASON FOR REVISION / CHANGE	DATE	BY	CHK	APP	CATEGORY:			

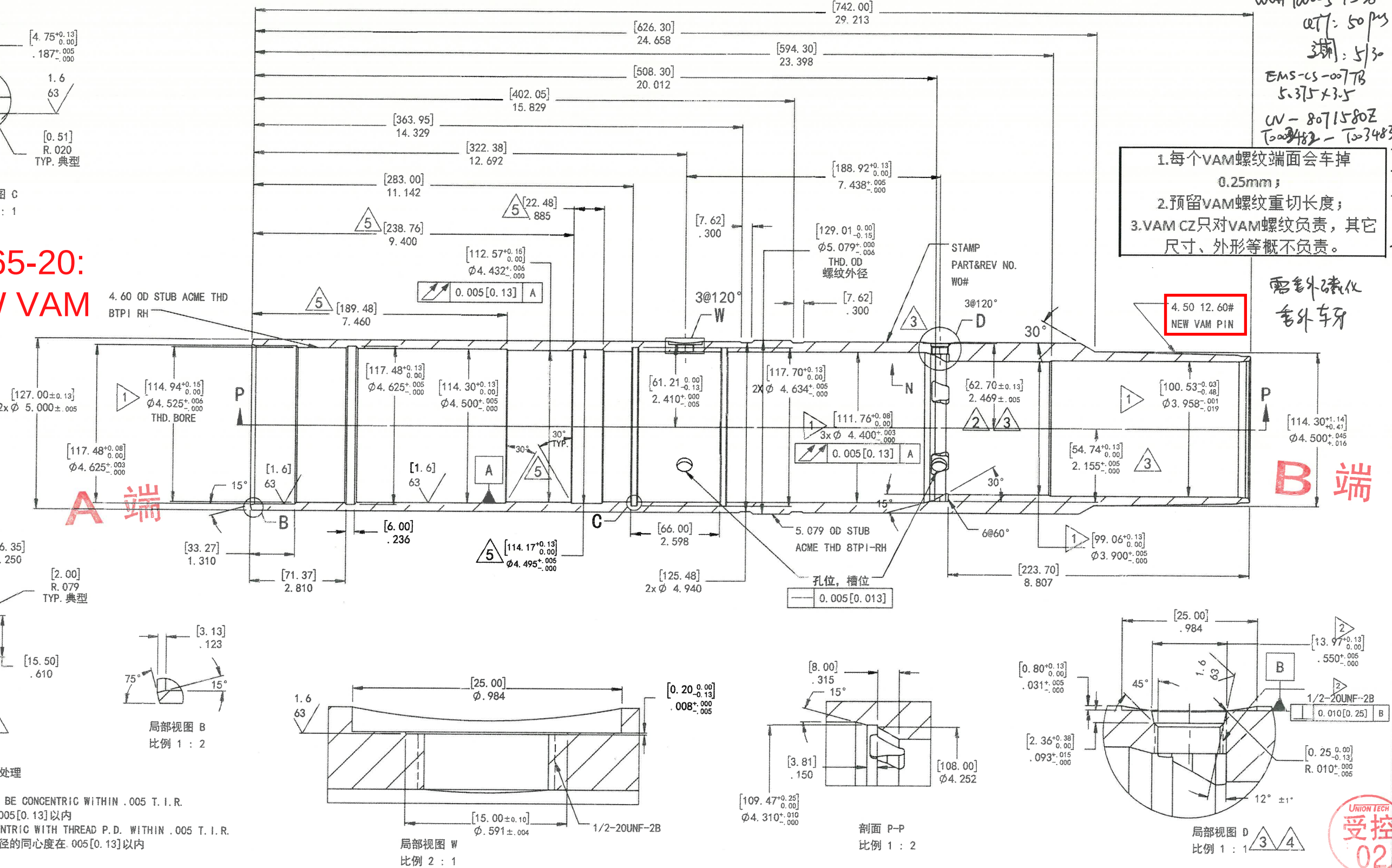
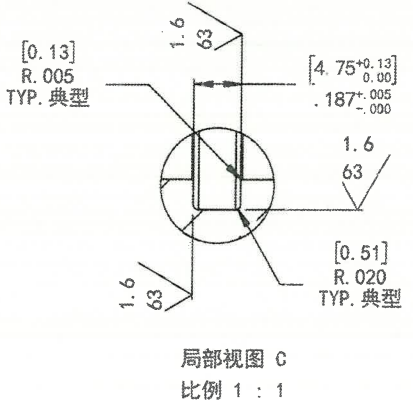
Watt Two 54538
WT: 50 pms
351: 5/30
EMS-US-007TB
5.375 x 3.5
UN-8071580Z
T003482 - T003483

٣٤٨٥
٣٤٨٦
٣٤٨٩
٥
٣٤٩١

需外磷化
需外车牙

1.每个VAM螺纹端面会车掉 0.25mm； 2.预留VAM螺纹重切长度； 3.VAM CZ只对VAM螺纹负责，其它 尺寸、外形等概不负责。
--

8121005465-20:
PIN端NEW VAM



技术要求:1. 去毛刺;2. 磷化处理

1 THESE DIMENSIONS TO BE CONCENTRIC WITHIN .005 T. I. R.
 这些尺寸的同心度在 .005[0.13] 以内

2 0.550 SHALL BE CONCENTRIC WITH THREAD P.D. WITHIN .005 T. I. R.
 0.550 [13.97] 和螺纹中径的同心度在 .005[0.13] 以内

Unit in [X] is mm.

5	优化设计, 取消UN牙	2026. 4. 10	李富全			TOLERANCE FOR MACHINED SURFACES UNLESS OTHERWISE SPECIFIED .X" = ± .06" X (mm) = ± 1.52mm .XX" = ± .03" X (mm) = ± 0.76mm .XXX" = ± .015" XX (mm) = ± 0.38mm X° = ± 2° X° = ± 2° ANGLES TO BE 30° FROM HORIZONTAL CONCENTRICITY .015" [0.38mm] TIR ALL EDGES .015"x45° [0.38mmx45°] INSIDE CORNERS .03"-.06" [0.8-1.5mm] R SURFACE FINISH $\sqrt{6.3}$ MAX. THREADS $\sqrt{3.2}$ MAX. CATEGORY:  	TITLE : BOTTOM SUB			DRAWING/MATERIAL NO.: 314-4500-C1-0006	HEAT TREAT: N/A	REV 05
4	增加角度公差, 增加同心度和垂直度要求	2025. 3. 19	李富全				PRODUCT FAMILY: 314			COATING: UTCS-001	PAGE 1/1	
3	优化设计, 尺寸变更	2025. 2. 28	李富全				MATERIAL : UTMS-C-001/LOW ALLOY CS 22 HRC 80 KS1			UNION TECH THIS DRAWING EMBODIES A CONFIDENTIAL PROPRIETARY DESIGN ORIGINATED BY UNION TECH CO., LLC. ALL DESIGN, MANUFACTURING, REPRODUCTION, USE AND SALES RIGHTS REGARDING THE SAME ARE EXPRESSLY RESERVED. THE RECIPIENT AGREES BY ACCEPTING THIS DRAWING NOT TO SUPPLY OR DISCLOSE ANY INFORMATION REGARDING IT TO AN UNAUTHORIZED PERSON OR CORPORATION * ALL PATENT RIGHTS ARE EXPRESSLY RESERVED BY UNION TECH CO. LLC. *		
2	Added 2.390, Added tolerance and concentricity requirements	2024. 11. 14	李富全				DATE: 2025/2/28					
1	Updated the coating	2020. 7. 28	YC				DRAWN BY: 李富全 CHECKED: 吴松勃 APPROVED: 王日华					
NO.	REASON FOR REVISION / CHANGE	DATE	BY	CHK	APP							

PURCHASE ORDER

太仓优尼泰克精密机械有限公司 Union Tech Co.,LLC					合同号: TPO042524/TPO042522			
					日期: 2026.6.11			
TO			SHIP TO			TERMS		
瓦卢瑞克石油天然气特殊设备（中国）有限公司 中国北京市朝阳区建国门外大街甲24号东海中心301室；100022 ATTN: 熊向辉 电话: +86 59233056			发货地址 江苏省常州市新北区新民东路18号 213033 物流部 卞耀娣 联系电话: 13685246880			交货方式: 自提（买方自己负责） 付款方式: 发货前付清全款 试刀废料: 不需提供类似产品的试刀废料		
Item 项目	Quantity 数量	Unit 单位	Part Number 产品件号	Drawing Number 产品图号	Description 产品描述	Unit Price 单价	Total Value 总额 含税	Requested delivery date 要求交付日期
1	50	PCS	TW0054271	314-4500-C1-0014	4.50 12.6# NEW VAM BOX			7/12
2	50	PCS	TW0054538	314-4500-C1-0006	4.50 12.6# NEW VAM PIN			7/12
Total:								
要求: 车牙+磷化+牙套								
Seller 供应商: 瓦卢瑞克石油天然气特殊设备（中国）有限公司					Buyer 买方: 太仓优尼泰克精密机械有限公司			
Seller Signature					Signature:			
Date					Date			

03/17/2025

Dear Sirs,

Please note that we (Innovex Downhole Solutions) are supplying the Goods requested from Union Tech Manufacturing for a contract with **Saudi Aramco** PO# 163288-00 . Below is the Catalogue description for the goods in the Saudi Aramco system. POs are issued via Aramco's system and Innovex needs to deliver with respect to Aramco's Contract.

If you have any questions concerning this, please contact our representative in **Saudi Arabia** below, contact information is available in the signature.

Vendor No.	10052108
Product Name	FULL-BORE MASTERFLOW ICDs
Product Number	T0002564 / 1001171549
Quantity Ordered	50 Tools

Material Number: 1001171549

SULTAN.ALQAHTANI.25

PO Text

SCREEN; FULL-BORE MASTERFLOW-SSD-ICD TYPE, SIZE 4.5", NEW VAM BXP, GRADE L-80, 4.5" 12.6# FULL-BORE MASTERFLOW-SSD-ICD, DRILL CHECK VALVE OR CAGE CHECK VALVE OPTION, NEW VAM BXP, GRADE L-80, OD 5.550 INCHES, TOP AND BOTTOM SUB ID 3.970 INCHES, SLEEVE ID 3.915 INCHES, BURST 5,758 PSI, COLLAPSE 5,544 PSI, TENSILE STRENGTH 217,700 LBS, TEMPERATURE RATING 275 F.

Best regards,



Abdulhadi Jaara

Senior Account Manager

Innovex-Inc

Mobile: +966-5000-74559

Email: Abdulhadi.Jaara@Innovex-inc.com

Orders information 订单信息		WI-TEC-001-09
SAP data 基本信息	Check item/确认项目	Product Type 产品类型 附件
	Size*OD * T*grades 尺寸*直径*壁厚*鋼級	4.50 12.6#NEW VAM BOX 4140
	Length Range/长度范围	
	Threading Type/车纹类型	NEW VAM BOX
	Customer /客户	太仓优尼泰克精密机械有限公司
	Delivery date/期望交货日	2026/07/12
	Production order quantity 需要生产数量	50
	Production order number 生产订单号	215198
	Material Number /物料号码	71001517
	Authorization Number 授权号码	1145551
	SO No. / Item 销售单号/项目	8121005465-10
	Customer order No/合同号	TPO042524/TPO042522
	PSL level/产品技术规格等级	
	Customer specification: 客户要求	TW0054271 图纸
	Bonded/保税管理	
Raw Material 原材料信息	Supplier/供应商	客户来料
	PO number 采购号	
protector 保护套 信息	Quantity delivered/交货数量	50
	Type/类型	NEW VAM BOX
Package 包装信息	Customer Requirement 客户 要求	客户来料包装,若无, 按瓦姆标准。

Orders information 订单信息		WI-TEC-001-09
SAP data 基本信息	Check item/确认项目	Product Type 产品类型 附件
	Size*OD * T*grades 尺寸*直径*壁厚*鋼級	4.50 12.6# NEW VAM PIN` 4140
	Length Range/长度范围	
	Threading Type/车纹类型	NEW VAM PIN
	Customer /客户	太仓优尼泰克精密机械有限公司
	Delivery date/期望交货日	2026/07/12
	Production order quantity 需要生产数量	50
	Production order number 生产订单号	215199
	Material Number /物料号码	71001517
	Authorization Number 授权号码	1145552
	SO No. / Item 销售单号/项目	8121005465-20
	Customer order No/合同号	TPO042524/TPO042522
	PSL level/产品技术规格等级	
	Customer specification: 客户要求	TW0054538 图纸
	Bonded/保税管理	
Raw Material 原材料信息	Supplier/供应商	客户来料
	PO number 采购号	
protector 保护套 信息	Quantity delivered/交货数量	50
	Type/类型	NEW VAM PIN
Package 包装信息	Customer Requirement 客户 要求	客户来料包装,若无，按瓦姆标准。

03/17/2025

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Product Number	T0002564 / 1001171549
Quantity Ordered	50 Tools

Material Number: 1001171549

SULTAN.ALQAHTANI.25

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Best regards,



Abdulhadi Jaara

Senior Account Manager

Innovex-Inc

Mobile: +966-5000-74559

Email: Abdulhadi.Jaara@Innovex-inc.com

PURCHASE ORDER

太仓优尼泰克精密机械有限公司 Union Tech Co.,LLC						合同号:		TPO042524/TPO042522	
						日期		2026.6.11	
TO				SHIP TO				TERMS	
瓦卢瑞克石油天然气特殊设备（中国）有限公司 中国北京市朝阳区建国门外大街甲24号东海中心301室；100022 ATTN: 熊向辉 电话: +86 59233056				发货地址 江苏省常州市新北区新民东路18号 213033 物流部 卞耀娣 联系电话: 13685246880		交货方式: 自提（买方自己负责）			
						付款方式: 发货前付清全款			
						试刀废料: 不需提供类似产品的试刀废料			
Item 项目	Quantit y 数量	Unit 单位	Part Number 产品件号	Drawing Number 产品图号	Description 产品描述	Unit Price 单价	Total Value 总额 含税	Requested delivery date 要求交付日期	
1	50	PCS	TW0054271	314-4500-C1-0014	4.50 12.6# NEW VAM BOX			7/12	
2	50	PCS	TW0054538	314-4500-C1-0006	4.50 12.6# NEW VAM PIN			7/12	
Total:									
要求: 车牙+磷化+牙套									
Seller 供应商: 瓦卢瑞克石油天然气特殊设备（中国）有限公司					Buyer 买方: 太仓优尼泰克精密机械有限公司				
Seller Signature					Signature:				
Date					Date				



QUALITY CERTIFICATE FOR STEEL PRODUCT OF HUBEI XINYEGANG STEEL CO., LTD.

湖北新冶钢有限公司钢材产品合格证明书

Our company's QMS is ISO9001:2008&ISO/TS16949:2009&AS9100C certified by Germany TÜV

本公司通过德国TÜV公司ISO9001:2008&ISO/TS16949:2009&AS9100C质量管理体系认证



卡片号/Batch No.

用户/Customer

T218035390

太仓优尼泰克精密机械有限公司/

技术标准/Technical Spec. No.

合同号/Contract No.

炉号/Heat No.

材型/Shape

加工用途/Designated Further

XYGN0694-2011B

Y812422001

7027162Z

无缝钢管/seamless steel tube

热压力加工/thermal pressure processing

产品名称/Description of Goods

钢号/Steel Grade

交货状态/Delivery Condition

规格/Size

合金结构钢/alloy structure steel

4140 80KSI

调质/quenched and tempered

136.53*23.81mm

特殊要求/Special Requirement:

I. 化学成分/Chemical Composition

1. 熔炼/Cast Analysis. %

C | 0.40 Si | 0.24 Mn | 0.94 P | 0.006 S | 0.006 Cr | 0.93 Ni | 0.03 Mo | 0.21 Cu | 0.02

2. 成品成分/Product Analysis %

1) C | 0.42 Si | 0.24 Mn | 0.94 P | 0.010 S | 0.005 Cr | 0.93 Ni | 0.02 Mo | 0.22 Cu | 0.02

2) C | 0.42 Si | 0.24 Mn | 0.94 P | 0.010 S | 0.005 Cr | 0.93 Ni | 0.02 Mo | 0.22 Cu | 0.02

II. 力学性能/Mechanical Properties

1. 拉伸/tensile 试验温度/test temp: 室温/R.M Temp 试验方向/test direction: 纵向/longitudinal 试验标准/test standard: ASTM A370

拉伸直径(mm)

| 12.62

| 12.60

屈服强度Rp0.5(MPa)

| 605

| 611

屈服强度Rp0.2(MPa)

| 605

| 607

Tensile specimen dia. (mm)

yield strength Rp0.5(MPa)

yield strength Rp0.2(MPa)

抗拉强度Rm(MPa)

| 727

| 726

屈服比Rp0.2/Rm

| 0.83

| 0.84

断后伸长率A(%)

| 27.0

| 27.0

tensile strength Rm(MPa)

yield ratio Rp0.2/Rm

elongation after fracture A(%)

断面收缩率 Z(%)

| 66

| 67

reduction of area Z(%)

2. 冲击/impact 试验温度/test temp: -29℃ 试验方向/test direction: 纵向/longitudinal 缺口深度/notch depth: 2mm 试验标准/test standard: ASTM E23 冲击尺寸/impact test specimen size: 10X10(mm)

夏比冲击吸收能量KV8(J)

| 109

| 106

| 123

charpy impact absorbed energy KV8(J)

3. 环状硬度/annular hardness 试验标准/test standard: ASTM E18

料号/sample No.

1

2

料号/sample No.

1

2

料号/sample No.

1

2

一象限内侧硬度1HRC

| 17.3

| 17.4

一象限内侧硬度2HRC

| 18.2

| 17.7

一象限内侧硬度3HRC

| 17.9

| 18.0

备注

/Remarks

判定日期/Judged on

捆数/Bundle(s)

支数/Piece(s)

重量/Weight(kg)

判定员/Judged by

质检部长/Quality Inspection Manager

2018-03-20

2

12

6482

明超/Ming Chao

刘光辉/Liu Guanghui

声明/Statement: 1. 本材料按照客户订单和上述标准要求进行生产、取样、测试以及检验。/The materials has been produced, sampled for testing and inspection in accordance with the requirements of customer orders and the above referred standard. 2. 有异议时, 来函请标明卡片号、钢号、炉号、规格、重量、发货日期、原因, 并将实物保管好。/If any dispute occurs, please advice the batch No., heat No., steel grade, size, weight, delivery date and reason for complain, and keep the material as it is. 3. 未经许可, 本产品未经加工不允许直接出口到其他国家和地区。/Without permission, the product without processing is not allowed to directly export to other countries and regions.



Page 1 of 3

通讯地址: 中国湖北省黄石市黄石大道316号, 质量检验部, 邮编: 435001, 电话: (0714) 6297272, 传真: (0714) 6297599
MAIL ADDRESS: QUALITY INSPECTION DEPARTMENT, No. 316 HUANGSHI AVENUE, HUANGSHI, HUBEI, CHINA. 435001, TEL: 86 714 6297

已审阅
龚正 18-04-03, 10:37



QUALITY CERTIFICATE FOR STEEL PRODUCT OF HUBEI XINYEGANG STEEL CO., LTD.

湖北新冶钢有限公司钢材产品合格证明书

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本公司通过德国TÜV公司ISO9001:2008&ISO/TS16949:2009&AS9100C质量管理体系认证



卡片号/Batch No.

用户/Customer

T218035390

太仓优尼泰克精密机械有限公司/

技术标准/Technical Spec. No.

合同号/Contract No.

炉号/Heat No.

材型/Shape

加工用途/Designated Further

XYGN0694-2011B

Y812422001

7027162Z

无缝钢管/seamless steel tube

热压力加工/thermal pressure processing

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规格/Size

合金结构钢/alloy structure steel

4140 80KSI

调质/quenched and tempered

136.53*23.81mm

No.1 quadrant inside hardness 1 HRC 一象限外侧硬度1HRC	16.5	16.5	No.1 quadrant inside hardness 2 HRC 一象限外侧硬度2HRC	16.8	16.6	No.1 quadrant inside hardness 3 HRC 一象限外侧硬度3HRC	17.2	17.1
No.1 quadrant outside hardness 1 HRC 一象限中心硬度1HRC	16.9	17.3	No.1 quadrant outside hardness 2 HRC 一象限中心硬度2HRC	17.5	16.9	No.1 quadrant outside hardness 3 HRC 一象限中心硬度3HRC	18.0	17.9
No.1 quadrant center hardness 1 HRC 一象限外侧硬度平均值HRC	16.8	16.7	No.1 quadrant center hardness 2 HRC 一象限内侧硬度平均值HRC	17.8	17.7	No.1 quadrant center hardness 3 HRC 一象限中心硬度平均值HRC	17.5	17.4
No.1 quadrant outside average hardness HRC			No.1 quadrant inside average hardness HRC			No.1 quadrant center average hardness HRC		

III. 微观检验/Microstructure

1. 晶粒度/grain size

晶粒度(级)
grain size(level)

9-8 9-8

IV. 无损检验/NDT

1. 超声波探伤 /UT |合格/good

2. 磁粉探伤 /MPI |合格/good

3. 剩余磁场(GS) /residual magnetic field(GS) |5

V. 热处理工艺/HT Process

1. 淬火加热温度(℃) |822-838

quenching temperature(℃)

淬火介质温度终点(℃) |22

finishing temperature of quench media(℃)

2. 回火温度(℃) |715-728

tempering temperature(℃)

3. 炉膛调查和检定时间 |2018-02-09

furnace hearth survey & calibration time

淬火保温时间(min)

quenching soaking time(min)

78

淬火冷却方法

quenching cooling method

|水冷/water cooling淬火介质温度开始(℃)

starting temperature of quench media(℃)

16

回火保温时间(min)

tempering soaking time(min)

148

回火冷却方法

tempering cooling method

|空冷/air cooling

炉膛调查和检定标准

furnace hearth survey & calibration standard

AMS-2750

备注
/Remarks

判定日期/Judged on	捆数/Bundle(s)	支数/Piece(s)	重量/Weight(kg)	判定员/Judged by	质检部长/Quality Inspection Manager
2018-03-20	2	12	6482	明超/Ming Chao	刘光辉/Liu Guanghui

声明/Statement: 1. 本材料按照客户订单和上述标准要求进行生产、取样、测试以及检验。/The materials has been produced, sampled for testing and inspection in accordance with the requirements of customer orders and the above referred standard. 2. 有异议时, 来函请标明卡片号、钢号、炉号、规格、重量、发货日期、原因, 并将实物保管好。/If any dispute occurs, please advice the batch No., heat No., steel grade, size, weight, delivery date and reason for complain, and keep the material as it is. 3. 未经许可, 本产品未经加工不允许直接出口到其他国家和地区。/Without permission, the product without processing is not allowed to directly export to other countries and regions.



Page 2 of 3

通讯地址: 中国湖北省黄石市黄石大道316号, 质量检验部, 邮编: 435001, 电话: (0714) 6297272, 传真: (0714) 6297599

MAIL ADDRESS: QUALITY INSPECTION DEPARTMENT, No. 316 HUANGSHI AVENUE, HUANGSHI, HUBEI, CHINA. 435001, TEL: 86 714 6297272, FAX: 86 714 6297599



QUALITY CERTIFICATE FOR STEEL PRODUCT OF HUBEI XINYEGANG STEEL CO., LTD.

湖北新冶钢有限公司钢材产品合格证明书

Our company's QMS is ISO9001:2008&ISO/TS16949:2009&AS9100C certified by Germany TÜV

本公司通过德国TÜV公司ISO9001:2008&ISO/TS16949:2009&AS9100C质量管理体系认证



卡片号/Batch No.

用户/Customer

T218035390

太仓优尼泰克精密机械有限公司/

技术标准/Technical Spec. No.

合同号/Contract No.

炉号/Heat No.

材型/Shape

加工用途/Designated Further

XYGN0694-2011B

Y812422001

7027162Z

无缝钢管/seamless steel tube

热压力加工/thermal pressure processing

产品名称/Description of Goods

钢号/Steel Grade

交货状态/Delivery Condition

规格/Size

合金结构钢/alloy structure steel

4140 80KSI

调质/quenched and tempered

136.53*23.81mm

4. 应力释放 stress relieve	YES
5. 矫直方式 straightening	旋转矫直/hot-straightening 热矫直完毕最低温度(℃) 396 min. temp. after hot straightening
	去应力方法 退火 S.R.method
	去应力温度(℃) 610-620 stress relieve temperature
	去应力保温时间(min) 148 stress relieve soaking time
VI. 尺寸, 外形和表面质量/Dimension, Appearance and Surface Quality	
尺寸、外形和表面质量/dimension, appearance and surface quality 合格/satisfactory	
交货硬度HBW/delivery hardness HBW 213-218	
VII. 其它类/Others	
1. 制造工艺/manufacturing process BOF+LF+RH+CC	
2. 压缩比/reduction ratio 25.8	

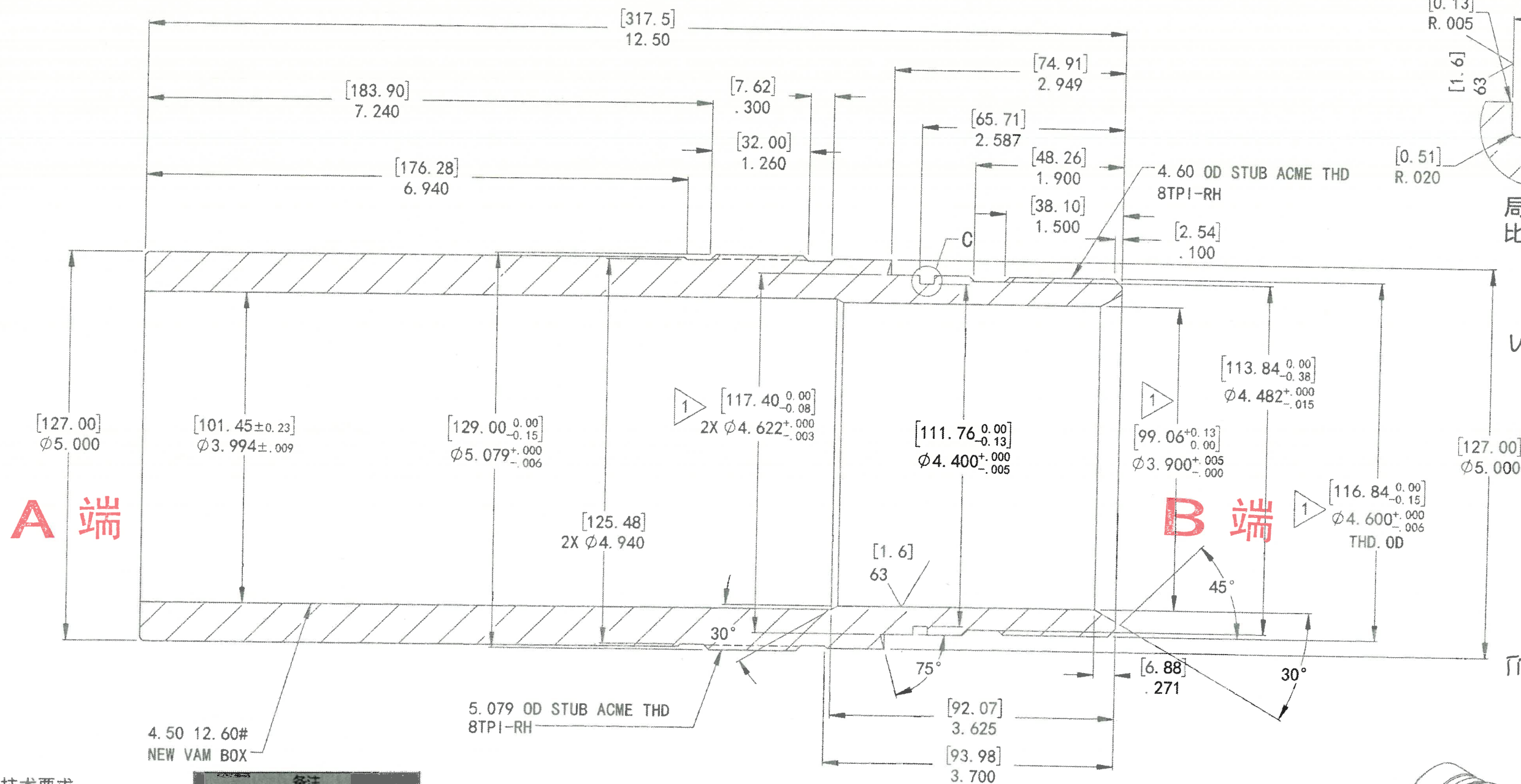
备注
/Remarks

判定日期/Judged on	捆数/Bundle(s)	支数/Piece(s)	重量/Weight(kg)	判定员/Judged by	质检部长/Quality Inspection Manager
2018-03-20	2	12	6482	明超/Ming Chao	刘光辉/Liu Guanghui

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局部视图 C
比例 2 : 1

Innovex part 163288-00
Item: 1

W# Tuo054271

QT: 50ps

交期: 5/30

EMS-CS-007TB (4140 80ksi)

5.375x3.5

4V-7027162Z

Tuo03112

Tuo03113

厂内喷砂+ 零件磷化+ 零件车牙
(UTCS-001, Rev0)

技术要求

1. 去毛刺
2. ACME螺纹处喷砂
3. 磷化处理

2

1 THESE DIMENSIONS TO BE CONCENTRIC WITHIN .005 TIR. 这些尺寸的同心率在.005[0.13]以内

Unit in [X] is mm.

备注

1. 每个VAM螺纹端面会车掉0.25mm;
2. 预留VAM螺纹重切长度;
3. VAM CZ只对VAM螺纹负责, 其它尺寸、外形等概不负责。

2

Added concentricity requirements

2024. 11. 14

李富全

1

Updated the coating

2020. 7. 28

YC

NO.

REASON FOR REVISION / CHANGE

DATE

BY

CHK

APP

CATEGORY:



TOLERANCE FOR MACHINED SURFACES UNLESS OTHERWISE SPECIFIED
 .X" = ± .06"
 .XX" = ± .03"
 .XXX" = ± .015"
 X° = ± 2°
 X.(mm) = ± 1.52mm
 X.(mm) = ± 0.76mm
 XX.(mm) = ± 0.38mm
 X° = ± 2°

ANGLES TO BE 30° FROM HORIZONTAL
 CONCENTRICITY .015"[0.38mm] TIR
 ALL EDGES .015"x45°[0.38mmx45°]
 INSIDE CORNERS .03"-.06"[0.8-1.5mm] R
 SURFACE FINISH 250/320 MAX. THREADS 125/320 MAX

TITLE: TOP SUB

MATERIAL:
L80 22HRC 80KSI

DATE: 2017/11/15

DRAWN BY:

JIM

CHECKED:

李富全

APPROVED:

陈楠

DRAWING/MATERIAL NO.:

314-4500-C1-0014

PRODUCT FAMILY:

314

HEAT TREAT:

N/A

COATING:

UTCS-001

REV

02

PAGE

1/1

UNION TECH

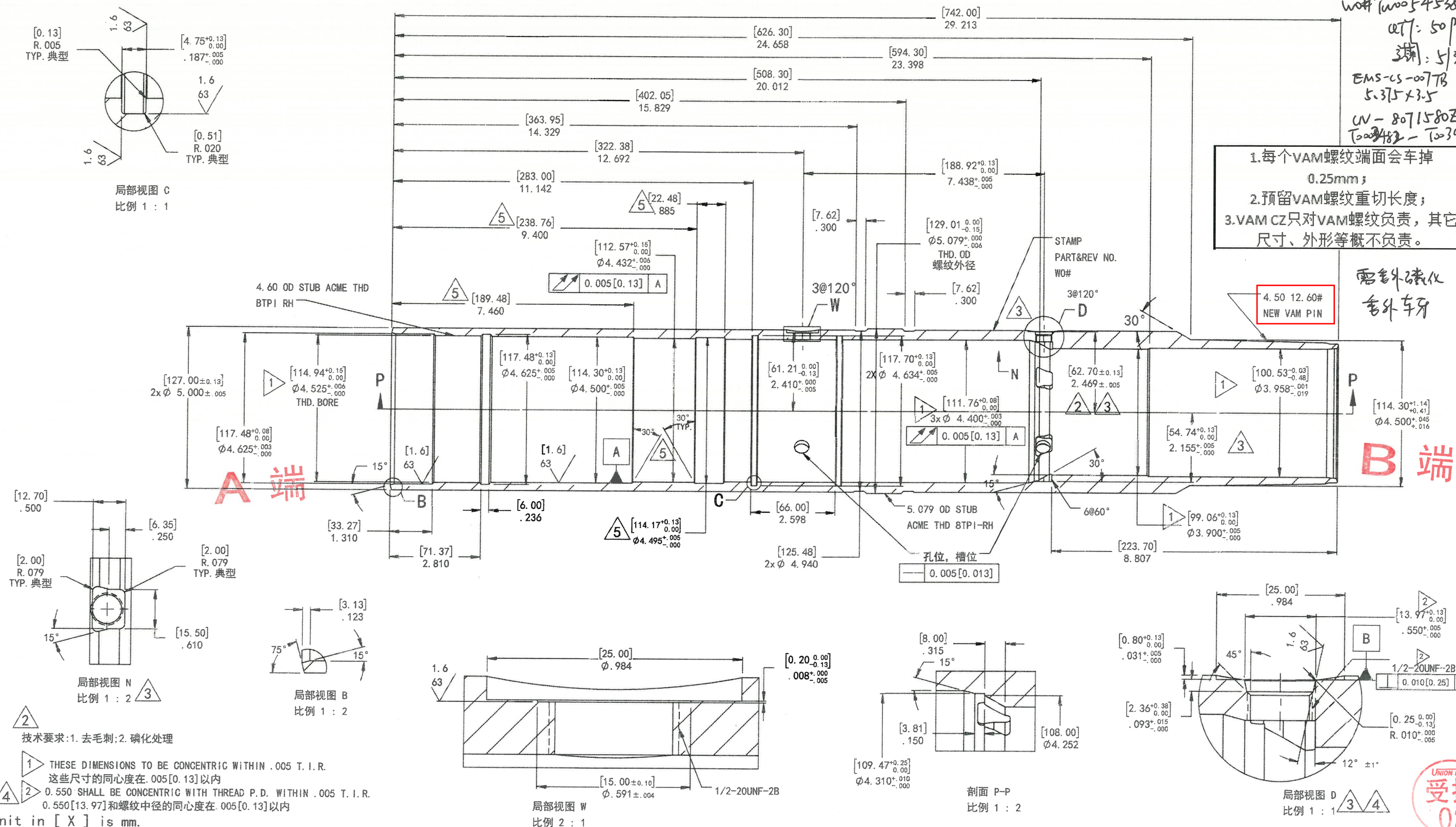
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UNION TECH
受控
02

W047 Two 54538
 alt: 50 pps
 3rd: 5/30
 EMS-US-007B
 5.375 x 3.5
 UN-8071580Z
 T003482 - T003483

T 3485
T 3487
T 3489
S
T 3491

需外磷化
需外车牙



UNION TECH
受控
02

5	优化设计，取消UN牙	2026. 4. 10	李富全			TOLERANCE FOR MACHINED SURFACES UNLESS OTHERWISE SPECIFIED .X" = ± .06" X (mm) = ± 1.52mm .XX" = ± .03" X (mm) = ± 0.76mm .XXX" = ± .015" X (mm) = ± 0.38mm X° = ± 2° X° = ± 2° ANGLES TO BE 30° FROM HORIZONTAL CONCENTRICITY .015"[0.38mm] TIR ALL EDGES .015"x45°[0.38mmx45°] INSIDE CORNERS .03"-.06"[0.8-1.5mm] R SURFACE FINISH ²⁵⁰ √ ^{1.3} MAX. THREADS ¹²⁵ √ ^{3.2} MAX	TITLE: BOTTOM SUB			DRAWING/MATERIAL NO.:	HEAT TREAT:	REV
4	增加角度公差，增加同心度和垂直度要求	2025. 3. 19	李富全				314-4500-C1-0006			N/A	05	
3	优化设计，尺寸变更	2025. 2. 28	李富全				MATERIAL:			PRODUCT FAMILY:	COATING:	PAGE
2	Added 2.390,Added tolerance and concentricity requirements	2024. 11. 14	李富全				UTMS-C-001/LOW ALLOY CS 22 HRC 80 KSI			314	UTCS-001 1	1/
1	Updated the coating	2020. 7. 28	YC				DATE: 2025/2/28			UNION TECH THIS DRAWING EMBODIES A CONFIDENTIAL PROPRIETARY DESIGN ORIGINATED BY UNION TECH CO. LLC. ALL DESIGN MANUFACTURING, REPRODUCTION, USE AND SALES RIGHTS REGARDING THE SAME ARE EXPRESSLY RESERVED. THE RECIPIENT AGREES BY ACCEPTING THIS DRAWING NOT TO SUPPLY OR DISCLOSE ANY INFORMATION REGARDING IT TO AN UNAUTHORIZED PERSON OR CORPORATION * ALL PATENT RIGHTS ARE EXPRESSLY RESERVED BY UNION TECH CO. LLC *		
NO.	REASON FOR REVISION / CHANGE	DATE	BY	CHK	APP	CATEGORY:	 	DRAWN BY:	CHECKED:	APPROVED:		
								李富全	吴机勃	王日华		



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技术标准/Technical Spec. No.

XYGN0694-2011B

合同号/Contract No.

Y821115001

炉号/Heat No.

8071580Z /

材型/Shape

无缝钢管/seamless steel tube

加工用途/Designated Further

热压力加工/thermal pressure processing

产品名称/Description of Goods

合金结构钢/alloy structure steel

钢号/Steel Grade

4140 80KSI /

交货状态/Delivery Condition

调质/quenched and tempered /

规格/Size

136.53*23.81mm /

特殊要求/Special Requirement: 7.56M定尺

I. 化学成分/Chemical Composition

1. 熔炼/Cast Analysis. %

C | 0.39 Si | 0.23 Mn | 0.93 P | 0.011 S | 0.005 Cr | 0.94 Ni | 0.14 Mo | 0.21 Cu | 0.04

2. 成品成分/Product Analysis %

1) C | 0.42 Si | 0.24 Mn | 0.92 P | 0.010 S | 0.004 Cr | 0.93 Ni | 0.14 Mo | 0.20 Cu | 0.04

2) C | 0.42 Si | 0.23 Mn | 0.93 P | 0.010 S | 0.005 Cr | 0.94 Ni | 0.14 Mo | 0.21 Cu | 0.04

II. 力学性能/Mechanical Properties

1. 拉伸/tensile 试验温度/test temp: 室温/R. M Temp 试验方向/test direction: 纵向/longitudinal 试验标准/test standard: ASTM A370

拉伸直径(mm)	12.75	12.78	屈服强度Rp0.5(MPa)	608	606	屈服强度Rp0.2(MPa)	606	595
Tensile specimen dia. (mm)			yield strength Rp0.5(MPa)			yield Strength Rp0.2(MPa)		
抗拉强度Rm(MPa)	704	705	屈强比Rp0.2/Rm	0.86	0.84	断后伸长率A(%)	28.5	29.0
tensile strength Rm(MPa)			yield ratio Rp0.2/Rm			elongation after fracture A(%)		
断面收缩率Z(%)	68	66						
reduction of area Z(%)								

2. 冲击/impact 试验温度/test temp: -29℃ 试验方向/test direction: 纵向/longitudinal 缺口深度/notch depth: 2mm 试验标准/test standard: ASTM E23 冲击尺寸/impact test specimen size: 10X10(mm)

夏比冲击吸收能量KV8(J)	141	142	143
charpy impact absorbed energy KV8(J)			

3. 环状硬度/annular hardness 试验标准/test standard: ASTM E18

料号/sample No.	1	2	料号/sample No.	1	2	料号/sample No.	1	2
一象限内侧硬度1HRC	19.1	19.2	一象限内侧硬度2HRC	19.4	19.3	一象限内侧硬度3HRC	19.1	19.4

备注
/Remarks

判定日期/Judged on

2018-04-25

捆数/Bundle(s)

2

支数/Piece(s)

9

重量/Weight(kg)

4454

判定员/Judged by

黄佳真/Huang Jiazhen

质检部长/Quality Inspection Manager

刘光辉/Liu Guanghui

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已审阅
龚正 18-05-06, 12:37



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T218049010

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太仓优尼泰克精密机械有限公司/

技术标准/Technical Spec. No.

XYGN0694-2011B

合同号/Contract No.

Y821115001

炉号/Heat No.

8071580Z

材型/Shape

无缝钢管/seamless steel tube

加工用途/Designated Further

热压力加工/thermal pressure processing

产品名称/Description of Goods

合金结构钢/alloy structure steel

钢号/Steel Grade

4140 80KSI

交货状态/Delivery Condition

调质/quenched and tempered

规格/Size

136.53*23.81mm

No.1 quadrant inside hardness 1 HRC 一象限外侧硬度1HRC	18.8	19.1	No.1 quadrant inside hardness 2 HRC 一象限外侧硬度2HRC	19.2	19.3	No.1 quadrant inside hardness 3 HRC 一象限外侧硬度3HRC	18.7	18.5
No.1 quadrant outside hardness 1 HRC 一象限中心硬度1HRC	18.4	19.0	No.1 quadrant outside hardness 2 HRC 一象限中心硬度2HRC	18.8	19.3	No.1 quadrant outside hardness 3 HRC 一象限中心硬度3HRC	18.9	19.5
No.1 quadrant center hardness 1 HRC 一象限外侧硬度平均值HRC	18.9	19.0	No.1 quadrant center hardness 2 HRC 一象限内侧硬度平均值HRC	19.2	19.3	No.1 quadrant center hardness 3 HRC 一象限中心硬度平均值HRC	18.7	19.3
No.1 quadrant outside average hardness HRC			No.1 quadrant inside average hardness HRC			No.1 quadrant centeraverage hardness HRC		

III. 微观检验/Microstructure

1. 晶粒度/grain size

晶粒度(级)

grain size(level)

|9-8

|9-8

IV. 无损检验/NDT

1. 超声波探伤 /UT |合格/good

2. 磁粉探伤 /MPI |合格/good

3. 剩余磁场(GS) /residual magnetic field(GS) |5

V. 热处理工艺/HT Process

1. 淬火加热温度(°C)

|817-834

淬火保温时间(min)

|44

淬火冷却方法

|水冷/water cooling 淬火介质温度开始(°C)

|10

quenching temperature(°C)

quenching soaking time(min)

quenching cooling method

start temp. of quench media(°C)

淬火介质温度终点(°C)

|14

finish temp. of quench media(°C)

2. 回火温度(°C)

|715-728

回火保温时间(min)

|87

回火冷却方法

|空冷/air cooling

tempering temperature(°C)

tempering soaking time(min)

tempering cooling method

3. 炉膛调查和检定时间

|2018-02-09

炉膛调查和检定标准

|AMS-2750

furnace hearth calibration time

furnace hearth calibration standard

备注

/Remarks

判定日期/Judged on	捆数/Bundle(s)	支数/Piece(s)	重量/Weight(kg)	判定员/Judged by	质检部长/Quality Inspection Manager
2018-04-25	2	9	4454	黄佳真/Huang Jiazhen	刘光辉/Liu Guanghui

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交货状态/Delivery Condition

调质/quenched and tempered

规格/Size

136.53*23.81mm

4. 应力释放 | YES
stress relieve
5. 矫直方式 | 自矫直/non-straightening
straightening

VI. 尺寸, 外形和表面质量/Dimension, Appearance and Surface Quality

尺寸、外形和表面质量/dimension, appearance and surface quality | 合格/satisfactory

交货硬度HBW/delivery hardness HBW | 212-222

VII. 其它类/Others

1. 制造工艺/manufacturing process | EAF+LF+RH+VD+CC

2. 压缩比/reduction ratio | 19.5

备注
/Remarks

判定日期/Judged on	捆数/Bundle(s)	支数/Piece(s)	重量/Weight(kg)	判定员/Judged by	质检部长/Quality Inspection Manager
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Orders information 订单信息		WI-TEC-001-09
SAP data 基本信息	Check item/确认项目	Product Type 产品类型 附件
	Size*OD * T*grades 尺寸*直径*壁厚*鋼級	4.50 12.6# NEW VAM BOX` 4140
	Length Range/长度范围	
	Threading Type/车纹类型	NEW VAM BOX
	Customer /客户	太仓优尼泰克精密机械有限公司
	Delivery date/期望交货日	2026/07/12
	Production order quantity 需要生产数量	50
	Production order number 生产订单号	215198
	Material Number /物料号码	71001517
	Authorization Number 授权号码	1145551
	SO No. / Item 销售单号/项目	8121005465-10
	Customer order No/合同号	TPO042524/TPO042522
	PSL level/产品技术规格等级	
	Customer specification: 客户要求	TW0054271 图纸
	Bonded/保税管理	
Raw Material 原材料信息	Supplier/供应商	客户来料
	PO number 采购号	
protector 保护套 信息	Quantity delivered/交货数量	50
	Type/类型	NEW VAM BOX
Package 包装信息	Customer Requirement 客户 要求	客户来料包装,若无，按瓦姆标准。